MILITARY SPECIFICATION

RACK, TEST TUBE, LABORATORY, WINTROBE, ENAMELED STEEL, 10 TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a laboratory test tube rack designed to accommodate ten Wintrobe hematocrit tubes in conjunction with determining sedimentation and hematocrit rates in blood samples.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, CODE ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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MILITARY

MIL-P-116	- Methods of Preservation-packaging.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24150	- Rack, Test Tube, Laboratory

(Copies of specifications, standards and drawing required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials shall be of the highest quality throughout, and as hereinafter specified.

3.2 Design and construction. The test tube rack shall be in strict conformance with DPSC Drawing No. 24150 and Bill of Materials for materials, design and dimensions. As indicated in the drawing, the test tube rack shall consist of a steel base with three leveling screw legs, a spirit level, an aluminum frame with a plastic back plate and a chromium plated brass cross bar with spring activated pressure buttons. The top and bottom bars of the aluminum frame shall have 10 triangular notches to accommodate 10 hematocrit tubes in a perpendicular position. The notches in the bottom bar shall be numbered consecutively from 1 to 10, as indicated in the drawing.

3.3 Performance. When tested as specified in 4.4.1, the base of the test tube rack shall be properly aligned, and the alignment of the cross bar and top and bottom bars of the frame shall permit the proper insertion and vertical positioning of the hematocrit tubes.

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3.4 Finish.

3.4.1 Aluminum components. All aluminum components shall be anodized.

3.4.2 Steel base. The steel base shall be coated with zinc-chromate primer and then finished with based-on hammertone enamel.

3.4.3 Brass components. The brass components shall be chrome-plated over nickel plating.

3.5 Identification marking. Each test tube rack shall be permanently and legibly marked with the following information·

Manufacturer's name or trademark,
Contract or purchase order number
"U.S."

Marking secured by adhesives shall not be used.

3.6 Workmanship. Workmanship shall be first class throughout All metal parts shall be free of corrosion and shall be throughly cleaned to remove foreign particles (e.g., dirt, grease, graphite, and metal chips) before final finish. The test tube racks shall be free of sharp edges, cracks, deformation, and other defects which detract from appearance or impair serviceability. The leveling screws and threaded members of the base shall be free of damage. The finish on component parts shall be free of discontinuitie and evidence of chipping and flaking.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4 1.1. Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties)of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105 and table I. The unit of product for examination shall be an individual test tube rack.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and table II. The unit of product for examination shall be an individual test tube rack.

TABLE II. For Test.

Characteristic	Requirement	Test	Inspection level	AQL
Performance	3.3	4.4.1	S-1	*

*Acceptance number shall be zero.

4.3 Examination. The test tube rack shall be examined to determine compliance with the requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III, but shall not be restricted to the classified possible defects listed in the table.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Rack not free from sharp edges and cracks.
102	Rack not free from deformation.
103	Rack not free from chips.
104	Rack not free from corrosion.
105	Coatings on rack not continuous and not free of evidence of chipping and flaking.
106	Threads on leveling screws and base not free of damage.
107	Leveling screws on base not capable of moving freely
108	Centers of tube holder and horizontal bars not properly aligned.
Minor	
201	Identification markings not complete and legible.
202	Rack not free of dirt and grease.

4.3.2 Dimensional examination. The racks shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Performance. A test tube rack shall be positioned on a horizontal surface. The three leveling screws shall be adjusted so that the spirit level indicates that the base of the test tube rack is horizontal. After leveling the base, ten hematocrit tubes, 5 millimeters (mm) in diameter, shall be inserted in the test tube rack.

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5. PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each rack shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each rack shall be packaged in accordance with MIL-P-116, method III. Box conforming to PPP-B-636, type CF, class domestic shall be utilized as a unit container.

5.1.2 Level C. Racks shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiveing activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Six (6) racks shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Six (6) racks shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged racks shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. National Stock Number.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6640-00-010-8358	RACK, TEST TUBE, LABORATORY, Wintrobe, Enameled Steel, 10 Tube

6.3 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians.
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
(DLA-DM)

Project No. 6640-1004

Reviewing activity

DoD - YB

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE
MIL-R-37624, RACK, TEST TUBE, LABORATORY, WINTROBE, ENAMELED STEEL, 10 TUBE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
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